ST. CLARET COLLEGE (AUTONOMOUS), BENGALURU
Department of Bachelor of Computer Applications (BCA)

VACATION MINI PROJECT REPORT

BIDPREMIER

IPL Fantasy Auction Simulator

Build your dream team. Outsmart AI rivals. Conquer the auction table.

Submitted By	**GOWTHAM**
Roll Number	**24BCA072**
Semester	**4th Semester**
Submission	**July 2026**

https://bid-premier-ipl.netlify.app/

CERTIFICATE

This is to certify that the mini project entitled

BIDPREMIER — IPL FANTASY AUCTION SIMULATOR

has been successfully developed and submitted by **GOWTHAM** (Roll No. **24BCA072**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **GOWTHAM**, Roll No. **24BCA072**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **"BIDPREMIER — IPL FANTASY AUCTION SIMULATOR"** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references have been duly cited.
- The conclusions drawn are based entirely on my own understanding.

Place: Bengaluru

Date: July 2026

Signature of the Student

GOWTHAM

Roll No: 24BCA072 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me during the development of this project.

I am grateful to the open-source developer community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which helped me understand complex JavaScript patterns, AI decision-making logic, real-time countdown implementation, and Chart.js integration.

I thank my family for their constant encouragement throughout the vacation period.

Building a real-time auction simulator with AI opponents — entirely in Vanilla JavaScript — was an immensely challenging and rewarding experience that pushed my understanding of the web platform significantly further.

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Roll No: 24BCA072 | 4th Semester, BCA

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ABSTRACT

Cricket is India's most-followed sport, and the IPL auction is one of the most exciting events in the cricketing calendar. BidPremier is an IPL Fantasy Auction Simulator that brings this experience to the browser, allowing a user to participate in a live-style auction against five AI-controlled franchise teams — Chennai Super Kings, Mumbai Indians, Royal Challengers, Sunrisers Hyderabad, and Kolkata Knight Riders — each with distinct bidding personalities and adaptive strategies.

The application simulates a complete IPL mega auction with a pool of 100 players, a ■100 Cr purse, real-time countdown timers, SOLD/PASS mechanics, live auction feeds, deep analytics, a player database, an achievements system, and three visual themes — all built using HTML5, CSS3, and Vanilla JavaScript, with all data persisted locally.

Key Features:

- Live auction engine with countdown timers, Going Once/Going Twice/SOLD! animations, and a dynamic bid feed.
- 5 AI franchise teams with unique bidding personalities that adapt dynamically to the auction state.
- 100-player pool spanning Batters, Bowlers, All-Rounders, and Keepers with individual ratings and base prices.
- Team Dashboard showing squad composition, purse usage, and player count for all 6 franchises.
- Analytics module: purse usage bar chart, role distribution doughnut, auction leaderboard, and top signings.
- Player Database with search and filter by role (All/Batters/Bowlers/All-Rounders/Keepers).
- Awards and Achievements system with XP levelling and unlockable trophies.
- Three themes: Dark Mode, Neon Mode, Stadium Mode — saved to localStorage.
- Configurable settings: timer duration, starting purse, and AI aggression level.
- Keyboard shortcuts for rapid bidding (1–5 = bid amounts, P = pause, Esc = pass).

Technologies Used:

HTML5, CSS3, Vanilla JavaScript, Chart.js, localStorage API. Deployed on Netlify.

TABLE OF CONTENTS

Certificate	2
Declaration	3
Acknowledgement	4
Abstract	5
Table of Contents	6
Chapter 1 – Introduction	7
Chapter 2 – Objectives of the Project	8
Chapter 3 – Existing System	9
Chapter 4 – Proposed System	10
Chapter 5 – Modules and Features	11
Chapter 6 – Technologies Used	15
Chapter 7 – Conclusion	17

INTRODUCTION

1.1 Background

The Indian Premier League auction is a high-stakes event watched by millions, where franchise owners bid for players under strict budget constraints. The strategic depth — managing a purse across 100 players, anticipating rival bids, balancing squad roles — makes it a compelling simulation subject. Fantasy sports platforms have popularised virtual team building, but none offer the real-time auction experience itself as an interactive browser game.

1.2 About the Project

BidPremier (v2.0) is a fully client-side IPL fantasy auction simulator. The user takes control of a franchise and competes against five AI teams in a simulated mega auction. The AI opponents are modelled with distinct personalities — CSK (Experience and Veterans), MI (Youth and Big Spenders), RCB (Aggressive Batting), SRH (Fast Specialists), KKR (Balanced Allrounders) — and adapt their bidding dynamically based on auction state.

Live at: **https://bid-premier-ipl.netlify.app/** — deployed on Netlify.

1.3 Scope

- Real-time auction simulation with timers, animations, and a live bid feed.
- AI-controlled franchise teams with adaptive bidding logic.
- 100-player database with ratings, averages, and base prices.
- Squad management, analytics, and team dashboard for all 6 franchises.
- Gamified achievements, XP levelling, and theme customisation.
- All data persisted locally — no server required.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of BidPremier:

1. To simulate a realistic IPL auction experience: Replicate the mechanics of a live IPL auction — countdown timers, Going Once/Going Twice/SOLD! announcements, pass options, and a dynamic bid feed — within the browser.

2. To implement AI franchise opponents with distinct personalities: Design five AI teams that bid differently based on their strategic profile (e.g., CSK favours veterans, MI prioritises youth), creating varied and unpredictable competition.

3. To build a comprehensive player database: Maintain a pool of 100 real-world IPL players with accurate role classifications, performance ratings, batting/bowling averages, and base prices.

4. To provide deep auction analytics: Offer visual insights into purse usage per team, role distribution in the user's squad, an auction leaderboard, and top signings to help users evaluate their auction performance.

5. To gamify the experience through achievements and XP: Motivate continued engagement through unlockable trophies, XP-based level progression, and notification pop-ups when milestones are reached.

6. To allow personalisation through themes and settings: Provide three visual themes and configurable auction parameters (timer, purse size, AI aggression) to let users tailor the experience.

7. To demonstrate advanced JavaScript programming ability: Implement a real-time game loop, AI decision logic, Chart.js visualisations, and localStorage-based state persistence using only Vanilla JavaScript.

EXISTING SYSTEM

3.1 Current Fantasy Sports Platforms

Existing platforms like Dream11, MPL, and My11Circle allow users to build fantasy teams based on real match performances, but none simulate the IPL auction itself as an interactive, real-time game. The following specific gaps exist:

a) No Real-Time Auction Simulation

Existing fantasy platforms use a draft or direct selection model. None replicate the live auction format with countdown timers, rival bidding, and budget pressure in real time.

b) No AI Opponent Logic

Current tools offer player selection against a static pool with no competitive bidding from AI entities. The strategic tension of outbidding rivals with limited funds is absent.

c) Server Dependency and Accounts

All major fantasy platforms require account creation, phone verification, and active internet connectivity. A lightweight, instant, browser-based alternative is not available.

d) No Educational Angle

Existing apps are designed for real-money or competitive gaming. BidPremier serves as an educational simulator for understanding auction strategy, purse management, and squad composition without monetary stakes.

3.2 Limitations Summary

- No real-time auction simulation with AI opponents in existing tools.
- Account and internet requirements restrict immediate access.
- No configurable difficulty or auction parameters.
- No integrated analytics or squad composition insights.
- No gamification beyond match-based points.

Chapter 4

PROPOSED SYSTEM

4.1 Overview

BidPremier proposes a fully browser-based, offline-capable IPL auction simulator that combines a real-time game engine, intelligent AI opponents, a rich player database, deep analytics, and a gamification layer — all accessible instantly without any account, installation, or server dependency.

4.2 Advantages

- **Real-Time Auction Engine:** Countdown timers, animated SOLD/PASS events, and a live bid feed create an immersive, pressure-filled auction experience.
- **Adaptive AI Opponents:** Five AI teams with distinct personalities and dynamic bidding logic provide varied, unpredictable competition across every game.
- **Instant Access:** No signup, no installation — open the URL and start an auction immediately.
- **Full Offline Capability:** All data persisted in localStorage; the game works after the first load without any network connection.
- **Configurable Difficulty:** Timer duration, starting purse, and AI aggression are all user-adjustable.
- **Educational Value:** Teaches purse management, role balancing, and auction strategy through simulated play.

4.3 Architecture

BidPremier is a single-page application with a persistent top navigation bar and seven view modules. A central JavaScript game engine manages the auction state machine (idle → running → sold/passed → next lot). AI decisions are evaluated every countdown tick. All state is serialised to localStorage after each transaction. Chart.js renders analytics charts dynamically from the persisted data.

MODULES AND FEATURES

BidPremier comprises seven modules accessible via the top navigation bar.

5.1 Home / Landing Page

The landing page establishes the application's identity with a dramatic full-screen dark background, the animated 'BID PREMIER' title in gold and teal gradients, and the tagline. Two call-to-action buttons — 'Start Mega Auction' and 'Browse Players' — provide immediate entry points. Four statistics counters display 100 Players, 6 Teams, 600 Total Purse Cr, and 18 Players Sold (updating live during an active auction). Below, six feature cards summarise the application's core capabilities.

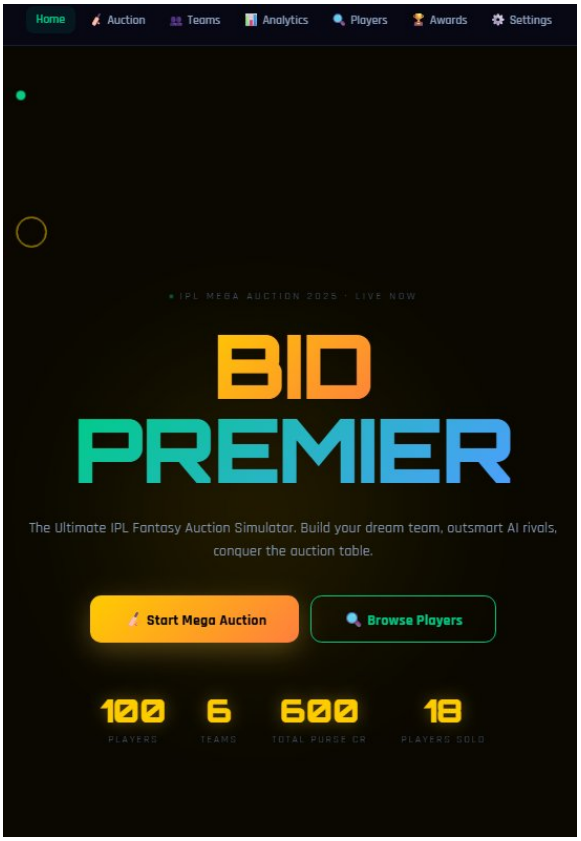


Figure 5.1 – BidPremier Landing Page

5.2 Auction Module

The Auction module is the centrepiece of the application. It renders the live auction interface with the current lot number, player card (name, nationality, role icon, rating, batting/bowling average, SR, matches, economy, popularity, recent form), current bid amount, highest bidder display, and a countdown timer.

Bidding Controls:

- **Bid increment buttons:** +25L, +50L, +1Cr, +2Cr, and ■ BID BIG +5Cr for aggressive bids.
- ■ **Pass:** Skip the current player and move to the next lot.
- ■ **Pause / ■ Next / ■ Restart:** Session controls.

AI Behaviour:

AI teams evaluate each player against their strategic profile and budget constraints. An aggressive AI may overbid on a target player; a conservative one may drop out early. This creates dynamic, unpredictable bid wars.

Live Auction Feed:

A scrollable feed on the right logs every bid, pass, and SOLD event in real time, with timestamps and team names, replicating the atmosphere of a live broadcast.

5.3 Teams Dashboard

The Team Dashboard displays all six franchises in a card grid. Each card shows the team name, personality tagline, strength score, purse remaining, players acquired, total spent, and squad slots with player name chips. The user's team is highlighted with a distinct border.



Figure 5.2 – Team Dashboard Showing All 6 Franchise Squad Cards

5.4 Analytics Module

The Analytics module provides four data visualisations:

- **Purse Usage by Team:** A grouped bar chart comparing spent vs remaining purse for all six franchises.
- **Role Distribution (Your Team):** A doughnut chart breaking down the user's squad by Batters, Bowlers, All-Rounders, and Keepers.
- **Auction Leaderboard:** A ranked table of all teams by players acquired, amount spent, remaining purse, average price, and strength score.
- **Top Signings / Sale Prices Over Time:** Charts tracking the most expensive acquisitions and bid price trends through the auction.

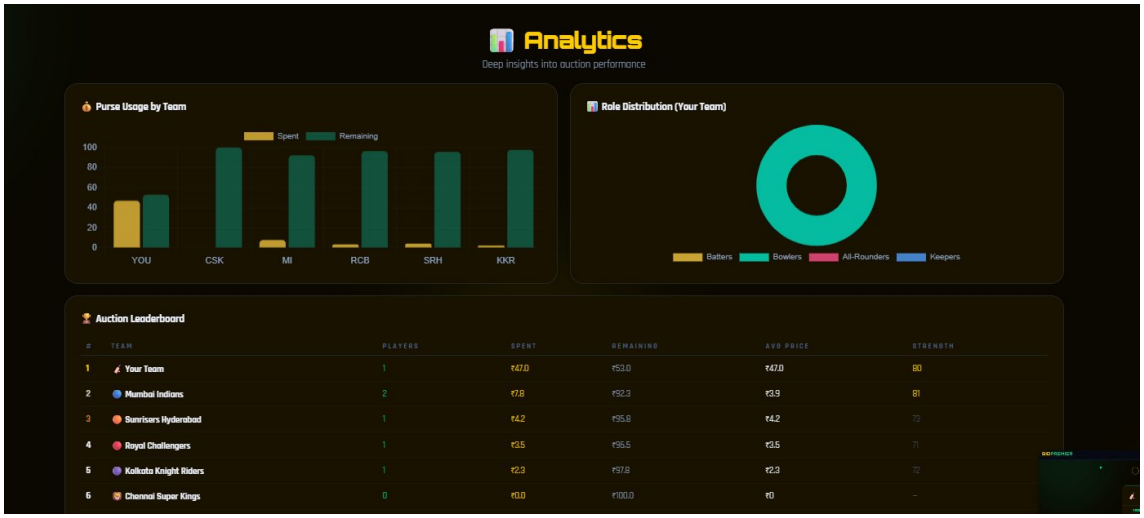


Figure 5.3 – Analytics Module with Purse Usage, Role Distribution, and Leaderboard

5.5 Player Database

The Player Database displays all 100 players in a responsive card grid. Each card shows the player's name, nationality, emoji icon, rating, batting average, bowling economy, base price, and availability status (Available/Sold). Filter tabs at the top (All/Batters/Bowlers/All-Rounders/Keepers) and a search bar enable quick lookup. Notable players include Virat Kohli, MS Dhoni, Rohit Sharma, KL Rahul, Jasprit Bumrah, and 95 others.

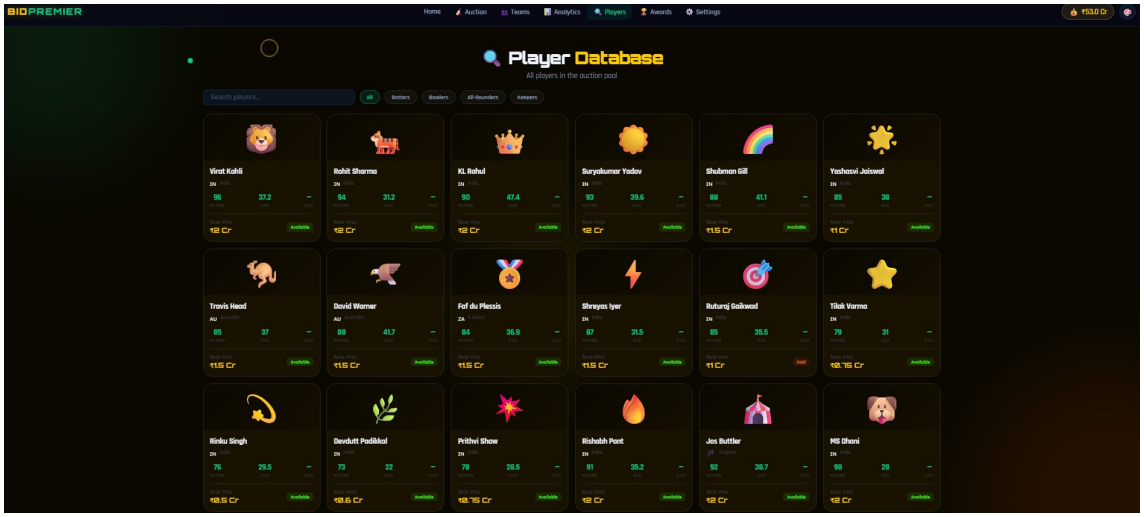


Figure 5.4 – Player Database with Role Filters and Player Cards

5.6 Awards and Settings Modules

Awards / Achievements:

Users earn XP and unlock achievement badges (e.g., Auction Master, Budget King) as they reach milestones. A level progression bar tracks XP earned toward the next rank. Achievement notifications pop up in real time during the auction.

Settings:

- **Theme:** Dark Mode (classic arena), Neon Mode (electric glow), Stadium Mode (warm lights).
- **Timer Duration:** 15 sec (Fast), 30 sec (Normal), 60 sec (Slow).
- **Starting Purse:** ■75 Cr, ■100 Cr, or ■120 Cr.
- **AI Aggression:** Easy, Normal, Hard.
- **Keyboard Shortcuts:** 1–5 for bid amounts, P to pause, Esc to pass.

Chapter 6

TECHNOLOGIES USED

BidPremier was developed using only native front-end web technologies. No framework or back-end infrastructure is used.

6.1 HTML5

HTML5 provides the structural foundation of all seven modules. Semantic elements, modal dialogs for auction events, and data attributes for player card rendering are used throughout. The application is structured as a single HTML file with dynamically toggled sections.

6.2 CSS3

CSS3 delivers the application's striking dark aesthetic with gold and teal gradients. Three complete theme systems (Dark, Neon, Stadium) are implemented using CSS Custom Properties, switching instantly via a class toggle on the root element. CSS animations power the SOLD! text, countdown pulse effects, and player card hover states. CSS Grid layouts the six-team dashboard.

6.3 JavaScript (ES6+, Vanilla)

The most complex component — the auction game engine — is implemented entirely in Vanilla JavaScript. Key systems include: the auction state machine (idle/running/sold/passed), a setInterval-based countdown timer with Going Once/Going Twice logic, AI decision functions evaluating bid ceiling per team personality, localStorage-based state serialisation after every event, dynamic player card rendering from a JSON data structure, and achievement unlock detection on every state change.

6.4 Chart.js

Chart.js renders the four analytics visualisations: a grouped bar chart for purse usage, a doughnut chart for role distribution, a line chart for sale prices over time, and a horizontal bar chart for top signings. Charts update dynamically after each lot is concluded.

6.5 localStorage API

The Web Storage API persists all auction state — current lot, team budgets, squad compositions, player availability, XP, achievements, and settings — to localStorage after every transaction. This enables the auction to survive page refreshes and resume seamlessly.

6.6 Netlify

The application is deployed as a static site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution.

#	Technology	Type	Purpose
1	HTML5	Markup	Structure, modals, player card templates
2	CSS3	Styling	Three themes, animations, grid layout
3	JavaScript ES6+	Programming	Auction engine, AI logic, state management
4	Chart.js	Library	Analytics charts (bar, doughnut, line)
5	localStorage	Browser API	Game state persistence
6	Netlify	Platform	Static hosting and CDN

CONCLUSION

BidPremier is one of the most technically ambitious projects that can be undertaken using only the front-end web platform. Building a real-time game engine with a functioning AI opponent system, a 100-player database, multi-chart analytics, a gamification layer, and three complete visual themes — all in Vanilla JavaScript with no framework or server — demonstrates a strong command of core web development concepts and a genuine passion for the subject matter.

The AI bidding system is particularly noteworthy. Modelling five distinct franchise personalities that respond dynamically to budget constraints and player desirability required careful design of decision functions and extensive testing to ensure competitive but not overwhelming opposition. This was, in effect, a practical exercise in basic game AI design.

The Chart.js analytics dashboard transforms raw auction data into actionable insights — purse efficiency, squad role balance, and competitive leaderboard rankings — adding a layer of strategic reflection that extends the value of each auction session.

Completing this project independently during the vacation period has been a deeply formative experience. The challenges of managing a complex stateful application without a framework, debugging AI decision logic, and ensuring smooth performance across a full 100-lot auction have significantly raised the bar for what I now expect from my own work.

BidPremier is a fully functional, publicly deployed IPL auction simulator that achieves all its stated objectives and stands as the strongest demonstration of front-end development skill produced during this academic year.

Live Project: https://bid-premier-ipl.netlify.app/